Genericity

Genericity

Intuitively:

- Observation: Regular economies have ``good properties.”
- Question: Are regular economies the norm or the exception?
- Goal: To show that *almost all economies are regular*.

Almost all economies

Preliminaries.

- For all $\ell \in \{1, 2, \dots, L\}$, let $a_\ell, b_\ell \in \mathbb{R}$.
- I is a rectangle in $\mathbb{R}^L$ if

$$I = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_L, b_L].$$

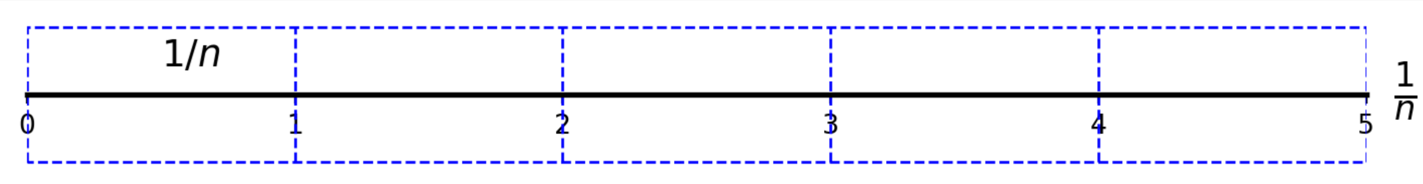
- The volume of I is $\mathrm{vol}(I) = \prod_{\ell=1}^{L} |b_\ell - a_\ell|$.

Lebesgue measure zero

- Let $A \subseteq \mathbb{R}^L$.
- A has Lebesgue measure zero, denoted by $\lambda(A) = 0$, if $\forall \epsilon > 0$, there is a countable collection of rectangles, $I_1, I_2, \ldots$ such that

$$\sum_{k=1}^{\infty} \mathrm{vol}(I_k) < \epsilon \text{ and } A \subseteq \bigcup_{k=1}^{\infty} I_k.$$

Example



- Let $A = [0, 5]$, $A \subset \mathbb{R}^2$.
- For each n, construct n rectangles I_n. Both horizontal sides of each I_n have length $\frac{b-a}{n}$. Both vertical sides of each I_n have length $1/n$.
- $(\forall n)\ A \subseteq \bigcup_{j=1}^{n} I_n$.
- And

$$(\forall \epsilon > 0)\,(\exists N)\ \left[n > N \implies \frac{b-a}{n} n \times \frac{1}{n} < \epsilon\right].$$

Some properties

- If A_n has Lebesgue measure zero for all $n = 1, 2, \ldots$, then $\bigcup_{n=1}^{\infty} A_n$ has Lebesgue measure zero.
- Let $P(x)$ be a proposition about the elements $x \in X \subseteq \mathbb{R}^L$.
- If $P(x)$ is true $\forall x \in X \setminus A$, and A has Lebesgue measure zero, we say that $P(\cdot)$ holds almost everywhere in X or that $P(\cdot)$ holds for almost all $x \in X$.

Topological notion of genericity

- Let $P(x)$ be a proposition about the elements $x \in X \subseteq \mathbb{R}^L$.
- $P(x)$ is generically true if $P(x)$ is true $\forall x \in G \subset X$, where G is an open and dense subset of X.
- Open. $\forall x \in G, \exists \epsilon_x > 0 : [y \in B(x, \epsilon_x) \implies y \in G]$.
- Dense. $\forall y \in X, \forall \epsilon > 0, \exists x \in G : y \in B(x, \epsilon)$.

Basic definitions

- Let z be homogeneous of degree zero.
- Normalize prices so that $p_L = 1$, and let $\hat{p} = (p_1, \ldots, p_{L-1}) \in \mathbb{R}^{L-1}_{++}$.
- Let $\hat{z} : \mathbb{R}^{L-1}_{++} \to \mathbb{R}^{L-1}$ be $\hat{z}(\hat{p}) = (z_1(\hat{p}, 1), \ldots, z_{L-1}(\hat{p}, 1))$.
- Let $F : \mathbb{R}^{L-1}_{+} \times \mathbb{R}^{LI}_{+} \mapsto \mathbb{R}^{L-1}$ be $F(\hat{p}, \omega) = \hat{z}(\hat{p})$ when the endowment assignment is ω.

Definition 1 The equilibrium correspondence $E : \mathbb{R}^{LI}_{+} \to \mathbb{R}^{L-1}_{++}$ is

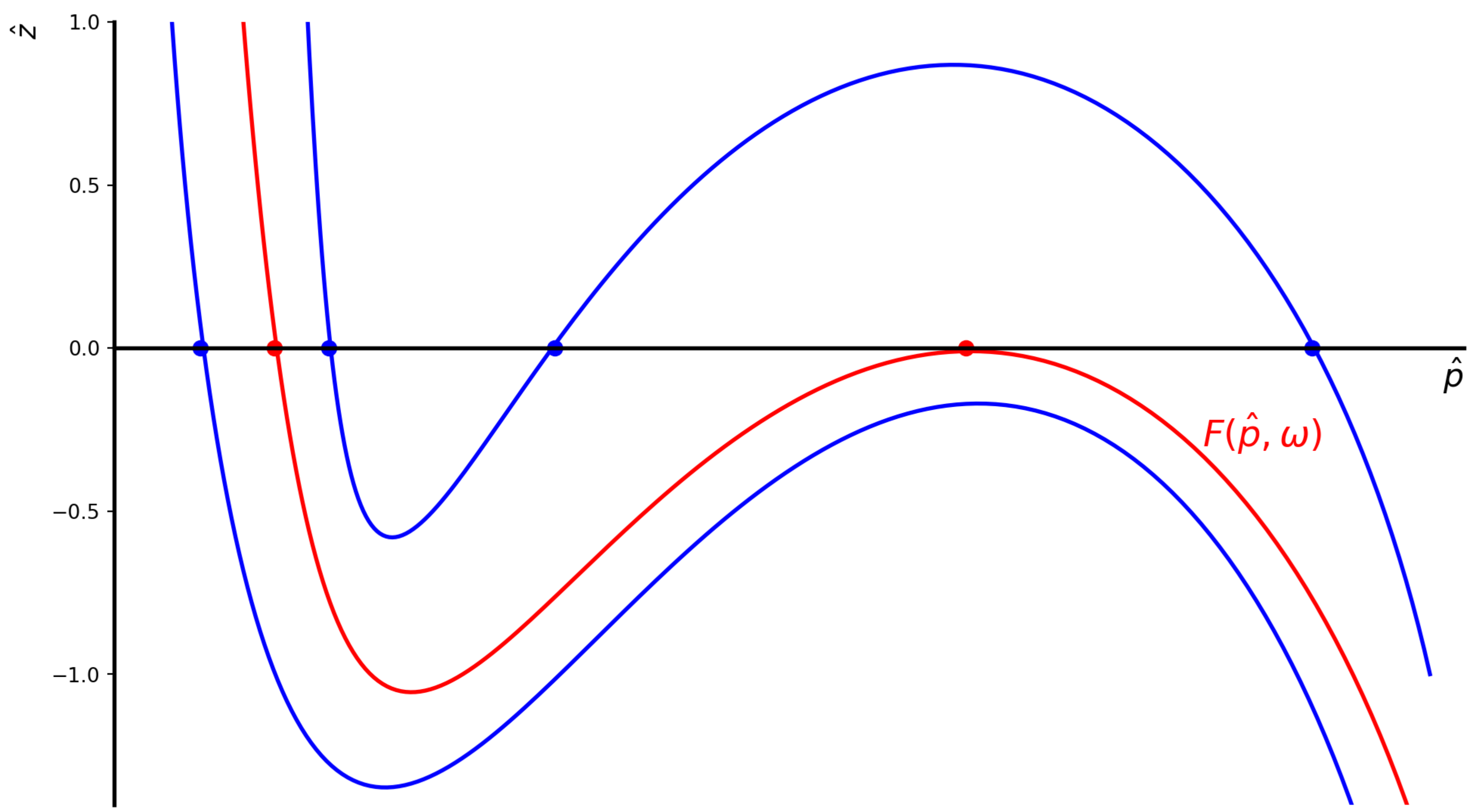
$$E(\omega) = \{\hat{p} \in \mathbb{R}^{L-1}_{++} : F(\hat{p}, \omega) = 0\}$$

Most economies are regular

Theorem 1 (Debreu) Let $\{\succeq\}_{i=1}^{I}$ be such that $D_i(p, \omega_i)$ is a C^1 function of (p, ω_i), and aggregate excess demand satisfies the hypotheses of the DGKN Lemma. Then there is a closed set $A \subset \mathbb{R}_{+}^{LI}$ of Lebesgue measure zero such that whenever $\omega' \in (\mathbb{R}_{+}^{LI} \setminus A)$

- the economy with preferences $\preceq_1, \dots, \preceq_I$ and endowment ω' is regular, so $E(\omega')$ is finite and odd, and
- if $E(\omega') = \{\hat{p}_1^*, \dots, \hat{p}_N^*\}$, then there is an open set $O \ni \omega'$, and C^1 functions $h_1, \dots, h_N$ such that

$$\omega \in O \implies E(\omega) = \{h_1(\omega), \dots, h_N(\omega)\}.$$



$\hat{z}$
1.0
0.5
0.0
−0.5
−1.0
$\hat{p}$
$F(\hat{p}, \omega)$